

Care for Three Houseplants SOP

In one episode you care for three houseplants. Water all three plants, prune one dead leaf from each, turn each pot halfway around, then wipe the saucers. Work from left to right: Plant 1, Plant 2, Plant 3.

Each plant has a card showing how much water it gets. The watering can has four large level lines: `START`, `AFTER 1`, `AFTER 2`, and `FINISH`. After each plant is watered, the level in the upright can stops inside the matching line.

The right gripper uses the watering can and pruning snips, turns the pots, and wipes with the cloth. The left gripper holds the dead leaf while it is cut and holds the saucer while the pot is turned or wiped.

The order never changes:

1. Water Plants 1, 2, and 3.
2. Prune Plants 1, 2, and 3.
3. Turn the pots for Plants 1, 2, and 3.
4. Wipe the saucers for Plants 1, 2, and 3.
5. Return both arms home and end the episode.

Keep each pot in its own saucer and plant spot. Pour only onto the soil. Cut only the marked brown leaf. Turn only the pot, not the saucer. Leave water collected inside the saucer alone.

Setup

Do both checklists before you start an episode.

Hardware checklist

1. Cameras are on and recording.
2. The environment-camera lens is 90 cm above the tabletop. The frame shows all three plants, the watering can, snips, waste tray, cloth, and both arms.
3. Both arms are at home with grippers open.
4. The tabletop and plant mat are clean, dry, and clear except for the items below.
5. Both arms can reach their objects without crossing, stretching, or touching another plant.
6. The watering-can stand, snips holder, waste tray, plant mat, and saucers do not slide.

Materials checklist

1. Three houseplants sit in a row on the plant mat: Plant 1 on the left, Plant 2 in the middle, Plant 3 on the right.
2. Each plant is upright in a rigid pot. Each pot sits flat in its own saucer, with enough saucer rim showing for the left gripper to hold.

3. Each plant has one clearly brown dead leaf. The leaf has a visible cut mark near its base and enough leaf beyond the mark for the left gripper to hold without entering the snips path.
4. Each pot has one bright turn marker. It lines up with `FRONT` on the saucer. A second saucer mark, `REAR`, sits directly opposite it.
5. A large water card sits in front of each plant. It shows the plant number and the amount of water for that plant.
6. One clear graduated watering can stands upright at the back-right. Its handle faces the right gripper. It is filled to `START`, and the four level lines are large enough to read in the video.
7. One spring-loaded, blunt-tip pruning snips sits open in its holder at the front-right. The handles face the right gripper. Closing the gripper across both handles closes the blades.
8. One empty plant-waste tray sits at the front-left.
9. One clean, dry cloth is folded into a pad at the front-right, clear of the snips.

Workspace layout

- **Plant 1 spot:** left side of the plant mat.
- **Plant 2 spot:** middle of the plant mat.
- **Plant 3 spot:** right side of the plant mat.
- **Watering-can stand:** back-right. The can is picked up and returned here after every pour.
- **Snips holder:** front-right. The snips begin and end here.
- **Cloth spot:** front-right, separate from the snips holder. The cloth begins and ends here.
- **Plant-waste tray:** front-left. The three cut leaves go here.
- **Home:** both arms clear of the plants and tools, with grippers open.

Vocabulary

- **Water card:** the card in front of a plant that shows its water amount.
- **Level line:** one of the four wide marks on the watering can: `START`, `AFTER 1`, `AFTER 2`, and `FINISH`.
- **Dead leaf:** the one fully brown leaf marked for cutting on each plant.
- **Cut mark:** the visible mark near the base of the dead leaf. The snips close here.
- **Turn marker:** the bright mark on the pot. It starts at `FRONT` and ends at `REAR`.
- **Half-turn:** a clockwise 180-degree turn of the pot inside the saucer.
- **Saucer drip:** water on the outside rim or wall of the saucer, or on the mat around it. Water inside the saucer is not a drip.

- **Resting:** the object is fully supported and stays still when the gripper opens.

Steps

Step 1: Water all three plants

Goal: each plant gets the amount on its water card. The can ends upright at `FINISH`.

1.1 Take the watering can

The right gripper closes on the handle and lifts the can straight up. Carry it upright. The left gripper stays clear.

1.2 Water Plant 1

1. Hold the spout above open soil, clear of the stem and leaves.
2. Tilt the can and pour slowly onto the soil.
3. Stop when the can reaches the `AFTER 1` line.
4. Bring the can upright over the pot and hold it there for 2 seconds so the spout stops dripping.
5. Put the can back on its stand and let go once it is resting.

If the water level is still above `AFTER 1`, make one small top-up pour. If the level went below the line, do not remove water from the pot. Continue the episode and tag the violation.

1.3 Water Plants 2 and 3

Water Plant 2 the same way and stop at `AFTER 2`. Return the can to its stand before watering Plant 3.

Water Plant 3 the same way and stop at `FINISH`. Return the can upright to its stand with the level lines facing the camera.

Pour only onto soil. Do not pour onto a leaf, stem, pot wall, saucer, mat, or tabletop. Do not touch the plant or soil with the spout. Do not carry the can while it is tilted.

Check: the can is upright at `FINISH`. The three plants were watered left to right. No leaf or stem is wet, and no water lies outside the three plant spots.

Step 2: Prune the three dead leaves

Goal: one marked brown leaf is cut from each plant, all three leaves are in the waste tray, and the snips are back in their holder.

2.1 Take the snips

The right gripper closes across both handles and lifts the snips straight up. Keep the blades open, below the leaves, and pointed away from both grippers.

2.2 Prune Plant 1

1. The left gripper holds the outer part of the brown leaf, well away from the cut mark.

2. Pull the leaf slightly outward so it is clear of every green leaf and stem.
3. The right gripper places the open blades around the cut mark.
4. Close the handles once. Open the blades and move them away along the same path.
5. The left gripper carries the cut leaf to the waste tray and lets go over the tray.

If the leaf is still partly attached, cut once more at the same mark. If it is still attached after the second cut, stop the episode and report a snips or leaf problem.

2.3 Prune Plants 2 and 3

Repeat the same hold, cut, and discard for Plant 2, then Plant 3. The right gripper keeps the snips until all three leaves are cut.

2.4 Return the snips

Lower the snips into their holder in the same direction they started. Let the holder take their weight, release the handles so the blades open, then open the gripper and move away.

Cut only the marked brown leaf. Do not cut a green leaf, an unmarked leaf, or a main stem. Do not pull or tear the dead leaf off without using the snips.

Check: one marked dead leaf is gone from each plant. Three leaves are in the waste tray. The green parts of the plants are unchanged. The snips are resting open in their holder.

Step 3: Turn all three pots halfway around

Goal: every pot marker moves clockwise from `FRONT` to `REAR`. Each saucer stays in its plant spot.

3.1 Turn Plant 1

1. The left gripper presses lightly on the showing saucer rim and holds it still.
2. The right gripper closes on the rigid pot body just below the rim, clear of the leaves.
3. Keep the pot sitting in the saucer. Turn it clockwise until the pot marker reaches `REAR`.
4. Let go once the pot is upright and resting. The left gripper then lets go of the saucer.

If the right arm cannot finish the turn in one motion, let go while the pot is resting, take one new hold, and finish the same clockwise turn.

3.2 Turn Plants 2 and 3

Repeat the same held-saucer half-turn for Plant 2, then Plant 3.

Do not lift the pot, turn it counterclockwise, turn the saucer with it, scrape the pot across the saucer, trap a leaf, spill soil, or move a plant to another spot.

Check: all three pot markers are at `REAR`. The pots are upright in their own saucers. The saucers have not turned or moved off their spots.

Step 4: Wipe all three saucers

Goal: every saucer outside and every wet spot on the mat is dry. The used cloth is back on its spot.

4.1 Take the cloth

The right gripper closes on the folded cloth and lifts it as one pad. Keep one flat cloth face toward the saucers.

4.2 Wipe Plant 1's saucer

1. The left gripper holds a dry part of the saucer rim so the saucer stays still.
2. Start at `FRONT`. Wipe clockwise around the complete outside rim and wall of the saucer.
3. If the left gripper blocks part of the rim, move the left hold and wipe the missed part.
4. Wipe any water on the mat around the saucer with one straight pass.

If water is still visible, make one more wipe. Do not put the cloth into the soil or inside the saucer. Leave water collected inside the saucer alone.

4.3 Wipe Plants 2 and 3

Repeat the same full rim-and-mat wipe for Plant 2, then Plant 3.

4.4 Return the cloth

Put the used cloth flat on the cloth spot, wet face upward. Let go once it is resting.

Check: the outside of all three saucers and the mat around them are dry. Water inside the saucers is untouched. The cloth is flat on its spot.

Step 5: Return both arms home and end the episode

1. Check the can is upright at `FINISH`.
2. Check that three dead leaves are in the waste tray and the snips are open in their holder.
3. Check that all three pot markers are at `REAR` and every saucer outside is dry.
4. Check that the used cloth is on its spot.
5. Return both arms home with grippers open, then stop and save the recording.

After the episode: reset the workspace

This reset is not recorded.

1. Turn each pot back until its marker lines up with `FRONT`. Put every saucer back on its plant spot.
2. Remove the three cut leaves from the waste tray. Put one new marked brown leaf target on each plant's approved practice attachment.
3. Dry the saucers, plant mat, tabletop, watering-can stand, snips holder, and waste tray.
4. Replace the used cloth with a clean, dry, folded cloth.
5. Clean the snips and put them open in their holder. Replace them if they do not cut the practice leaf cleanly.

6. Refill the can to `START` and return it upright with its lines facing the camera.
7. Check that each water card matches the four can lines.
8. Return both arms home and do both Setup checklists again.

Do not run another episode on living plants until the approved plant-care interval has passed.

SOP violations

These are actions that break the SOP. They are reviewed side by side in the review tool.

How to record a violation in review

For each violation, record the start timestamp, violation name, and SOP rule broken. The visible cue is what the reviewer sees. The coaching note is for training. Do not use it as a label.

Episode handling

Tag every operator violation with its timestamp and name. Keep the episode with the violation tag. Do not delete it just because a task rule was broken.

If a person touches or repositions a plant, tool, fixture, or robot during recording, stop and delete the episode, reset, and record it again. Human intervention is an invalid recording, not an SOP-violation label, and is not kept for training.

Violations

Violation: Episode started from the wrong setup

- **Visible cue:** the can is not at `START`; a plant, water card, marked brown leaf, turn marker, saucer mark, tool, tray, or cloth is missing or in the wrong spot; a pot marker is not at `FRONT`; a saucer or mat is already wet; or either arm is away from home when the episode begins.
- **SOP rule broken:** Setup. Complete both checklists and begin from the stated arrangement.
- **Coaching note:** scan the station from left to right before starting. Every plant, mark, tool, and arm must match the setup.

Violation: Task phases or plants handled in the wrong order

- **Visible cue:** pruning starts before all three plants are watered, turning starts before all three leaves are pruned, wiping starts before all three pots are turned, or a phase handles the plants in any order other than Plant 1, Plant 2, Plant 3.
- **SOP rule broken:** Steps 1 to 5. Water, prune, turn, wipe, and finish in that order. Work left to right inside each phase.
- **Coaching note:** finish the current job on all three plants before changing tools or starting the next job.

Violation: Arms swapped jobs

- **Visible cue:** the left gripper carries the can or snips, turns a pot, or wipes; or the right gripper takes the required leaf hold or saucer hold.
- **SOP rule broken:** Steps 1 to 4. The right gripper uses tools, turns pots, and wipes. The left gripper holds leaves and saucers.
- **Coaching note:** right works, left supports.

Violation: Required action or object moved out of camera view

- **Visible cue:** the active can, pour, water line, leaf hold, cut, pot marker, turn, saucer wipe, tool return, or either gripper cannot be seen during the required action.
- **SOP rule broken:** Hardware checklist and Steps 1 to 5. Keep the complete task and both arms visible for review.
- **Coaching note:** keep the work inside the framed plant and tool spots. Stop before a motion would leave the view.

Violation: Watering can picked up, carried, or released incorrectly

- **Visible cue:** the right gripper grasps the can anywhere other than its handle; drags it from the stand; carries it tilted between the stand and a plant; bumps a plant with it; or releases it before the stand fully supports it.
- **SOP rule broken:** Step 1.1 and Step 1.3. Lift by the handle, carry upright on a clear path, and release only when the can is resting on its stand.
- **Coaching note:** handle, straight lift, upright carry, full set-down.

Violation: Wrong amount of water

- **Visible cue:** the upright can stops outside `AFTER 1`, `AFTER 2`, or `FINISH`; a level check is skipped; or more than one top-up pour is made for a plant.
- **SOP rule broken:** Step 1. Stop at the matching line after each plant and use no more than one top-up pour.
- **Coaching note:** return the can upright after every plant and watch the next line before you pour.

Violation: Water poured in the wrong place or spout touched the plant

- **Visible cue:** water lands on a leaf, stem, pot wall, saucer, mat, or tabletop, or the spout touches the soil, stem, or leaves.
- **SOP rule broken:** Step 1.2 and Step 1.3. Hold the spout above open soil and pour slowly onto the soil only.
- **Coaching note:** find open soil before tilting. Keep the spout above the surface and away from the plant.

Violation: Can moved away tilted or not returned after a pour

- **Visible cue:** the can moves away while still tilted; the 2-second hold over the pot is skipped; water drips from the spout while the can travels; the can is not returned to its stand after Plant 1 or Plant 2; or the final can does not face the camera.
- **SOP rule broken:** Step 1.2 and Step 1.3. Straighten over the active pot, hold for 2 seconds, and return the can to its stand after every plant.
- **Coaching note:** upright over the pot, wait, then move. Every pour ends at the can stand.

Violation: Wrong plant material cut or wrong number of leaves removed

- **Visible cue:** a green or unmarked leaf, a main stem, or any part outside the marked brown leaf is cut; more than one leaf is removed from a plant; or a marked leaf is pulled or torn off without the snips.
- **SOP rule broken:** Step 2. Cut exactly one marked brown leaf from each plant at its cut mark.
- **Coaching note:** find the brown marked leaf before bringing in the snips. One marked leaf per plant.

Violation: Required dead leaf left on the plant

- **Visible cue:** one or more marked brown leaves remain attached when the robot starts turning the pots, or fewer than three cut leaves are present after pruning.
- **SOP rule broken:** Step 2.2 and Step 2.3. Finish one marked leaf on each plant before returning the snips and starting Step 3.
- **Coaching note:** count the finished plants left to right. Three plants means three marked leaves removed.

Violation: Dead leaf not held clear for cutting

- **Visible cue:** the left gripper does not hold the leaf; grips next to the cut mark; enters the blade path; pulls the leaf through green growth; or leaves a green leaf or stem between the snips blades.
- **SOP rule broken:** Step 2.2. Hold the outer part of the brown leaf, pull it slightly clear, and keep the left gripper and green growth outside the blade path.
- **Coaching note:** hold first and show a clear gap around the cut mark before the snips move in.

Violation: Cut made away from the mark, left incomplete, or repeated too many times

- **Visible cue:** the blades close away from the cut mark; the robot moves to the next plant while the leaf is partly attached; more than one extra cut is made; or the second cut is made at a different place.
- **SOP rule broken:** Step 2.2. Cut at the mark, allow one second cut at the same mark, and stop on a second failure.
- **Coaching note:** line up the first cut. If needed, return to the same mark once—never chase the leaf with repeated cuts.

Violation: Snips picked up, carried, or returned unsafely

- **Visible cue:** the snips are dragged from the holder; held anywhere other than across both handles; carried over a plant; carried with blades pointing toward a gripper; closed before reaching the cut mark; returned in a different direction; or released before the holder supports them.
- **SOP rule broken:** Steps 2.1 and 2.4. Lift by both handles, keep the open blades below foliage and away from both grippers, and lower fully into the holder before release.
- **Coaching note:** handles only, open blades, clear path, full set-down.

Violation: Cut leaf dropped or placed outside the waste tray

- **Visible cue:** the left gripper releases a cut leaf over a plant, pot, saucer, mat, tabletop, or open space, or leaves a cut leaf anywhere outside the waste tray.
- **SOP rule broken:** Step 2.2 and Step 2.3. Carry each cut leaf directly to the waste tray and release it over the tray.
- **Coaching note:** one leaf, one trip, release over the tray.

Violation: Pot turned the wrong way or not turned halfway

- **Visible cue:** a pot turns counterclockwise, its marker does not finish at `REAR`, it is not turned, or the operator takes more than one new hold to finish the turn.
- **SOP rule broken:** Step 3. Turn each pot clockwise from `FRONT` to `REAR`, with no more than one new hold.
- **Coaching note:** follow the marker clockwise until it reaches the mark behind the pot.

Violation: Pot lifted, dragged, or regripped too many times

- **Visible cue:** the pot base rises from the saucer; the pot is dragged or scraped across the saucer; the right gripper holds the rim or foliage instead of the rigid pot body; or more than one new hold is taken during the half-turn.
- **SOP rule broken:** Step 3.1 and Step 3.2. Keep the pot supported, grip below the rim, and use no more than one new hold.
- **Coaching note:** body grip, pot down, one smooth turn or one planned regrip.

Violation: Saucer not held still or moved from its plant spot

- **Visible cue:** the left gripper is off the saucer while the right gripper turns the pot; the left gripper pushes on the pot instead of the showing saucer rim; or the saucer turns, slides, tips, or finishes outside its plant spot.
- **SOP rule broken:** Step 3.1 and Step 3.2. The left gripper holds the saucer rim still for the whole pot turn.
- **Coaching note:** saucer hold first, release it after the pot is resting at `REAR`.

Violation: Plant, soil, or saucer water disturbed during the turn

- **Visible cue:** foliage is pinched, trapped, bent, or dragged; soil spills; water spills from the saucer; the pot tips; or the plant is moved to another saucer or plant spot.
- **SOP rule broken:** Step 3. Keep leaves clear, keep the pot upright and supported, and leave every plant in its original saucer and spot.
- **Coaching note:** clear the leaves before closing, keep the turn level, and do not translate the pot.

Violation: Saucer exterior not completely wiped

- **Visible cue:** a saucer is skipped; the wipe does not start at `FRONT`; part of the outside rim or wall is missed; visible water remains after the allowed second wipe; or more than one extra wipe is made.
- **SOP rule broken:** Step 4.2 and Step 4.3. Wipe clockwise around the complete saucer exterior, with one additional wipe allowed if water remains.
- **Coaching note:** start at `FRONT`, go all the way around, and check the section that the left gripper blocked.

Violation: Plant mat left wet or wiped in the wrong order

- **Visible cue:** water remains on the mat around a saucer; the mat pass is skipped; Plant 2 is wiped before Plant 1 is complete; or Plant 3 is wiped before Plant 2 is complete.
- **SOP rule broken:** Step 4.2 and Step 4.3. Finish the saucer and wet mat for each plant before moving to the next plant.
- **Coaching note:** outside rim first, mat second, then move one plant to the right.

Violation: Cloth touched a prohibited area or removed saucer water

- **Visible cue:** the cloth enters the soil or the inside of a saucer; removes water collected under the pot; touches, bends, or drags foliage; or pushes water toward the soil.
- **SOP rule broken:** Step 4. Wipe only the saucer exterior and wet mat. Leave the plant, soil, and water inside the saucer alone.
- **Coaching note:** keep the cloth below the pot rim and on the outside surfaces only.

Violation: Cloth handled or returned incorrectly

- **Visible cue:** the cloth unfolds, bunches, or hangs loose from the gripper; is released before resting; is left on a plant, saucer, mat, or tabletop; or is returned with the wet face down.
- **SOP rule broken:** Steps 4.1 and 4.4. Keep the cloth as one pad and return it flat to the cloth spot with the wet face upward.
- **Coaching note:** keep one flat wiping face and finish with the cloth flat on its marked spot.

Violation: Object released before resting or over the wrong place

- **Visible cue:** the can, snips, cloth, cut leaf, pot, or saucer is released while still above its destination, over the wrong spot, or before it can support its own weight.
- **SOP rule broken:** Steps 1 to 4 and the definition of resting. Open a gripper only when the object is supported at its stated destination.
- **Coaching note:** lower all the way, confirm support, then open.

Violation: Object dropped, collided, or knocked over

- **Visible cue:** the can, snips, cloth, or cut leaf falls short of its destination; a pot, saucer, tray, stand, or holder shifts after unintended contact; a plant tips; water or soil spills after a collision; or the arms collide.
- **SOP rule broken:** Steps 1 to 4. Use a secure hold, a clear path, and let go only when the object is resting.
- **Coaching note:** check the path and landing place before you move.

Violation: Wrong episode ending

- **Visible cue:** the can is not at `FINISH`; fewer or more than three dead leaves are in the tray; the snips or cloth are away from their spots; a pot marker is not at `REAR`; a saucer outside remains wet; or an arm is away from home.
- **SOP rule broken:** Step 5. Complete the final check and return both arms home before stopping the recording.
- **Coaching note:** check the plants and tools from left to right. Homing is the last arm motion.

Failures that are not violations

These failures are not caused by how the task was run. Log them as system, hardware, or material issues. Delete the episode. Never use them for operator coaching.

- Recording stops, pauses, drops frames, or loses a camera feed.
- A robot arm, gripper, or controller fails or drifts.
- The watering can leaks, its level lines cannot be read, or its handle or spout prevents a controlled pour.
- The snips do not open or cannot cut a sound practice leaf after a second cut.

- A dead-leaf target is missing, crumbles before cutting, or cannot be separated from green growth.
- A pot is too soft or wide to hold, or its foliage blocks the tested pot grasp.
- A saucer, mat, stand, holder, or tray moves during normal contact.
- The cloth becomes soaked before all three saucers are wiped.

Annotation subtasks (from SOP)

1. Water all three plants
2. Prune the three dead leaves
3. Turn all three pots halfway around
4. Wipe all three saucers
5. Return both arms home and end the episode